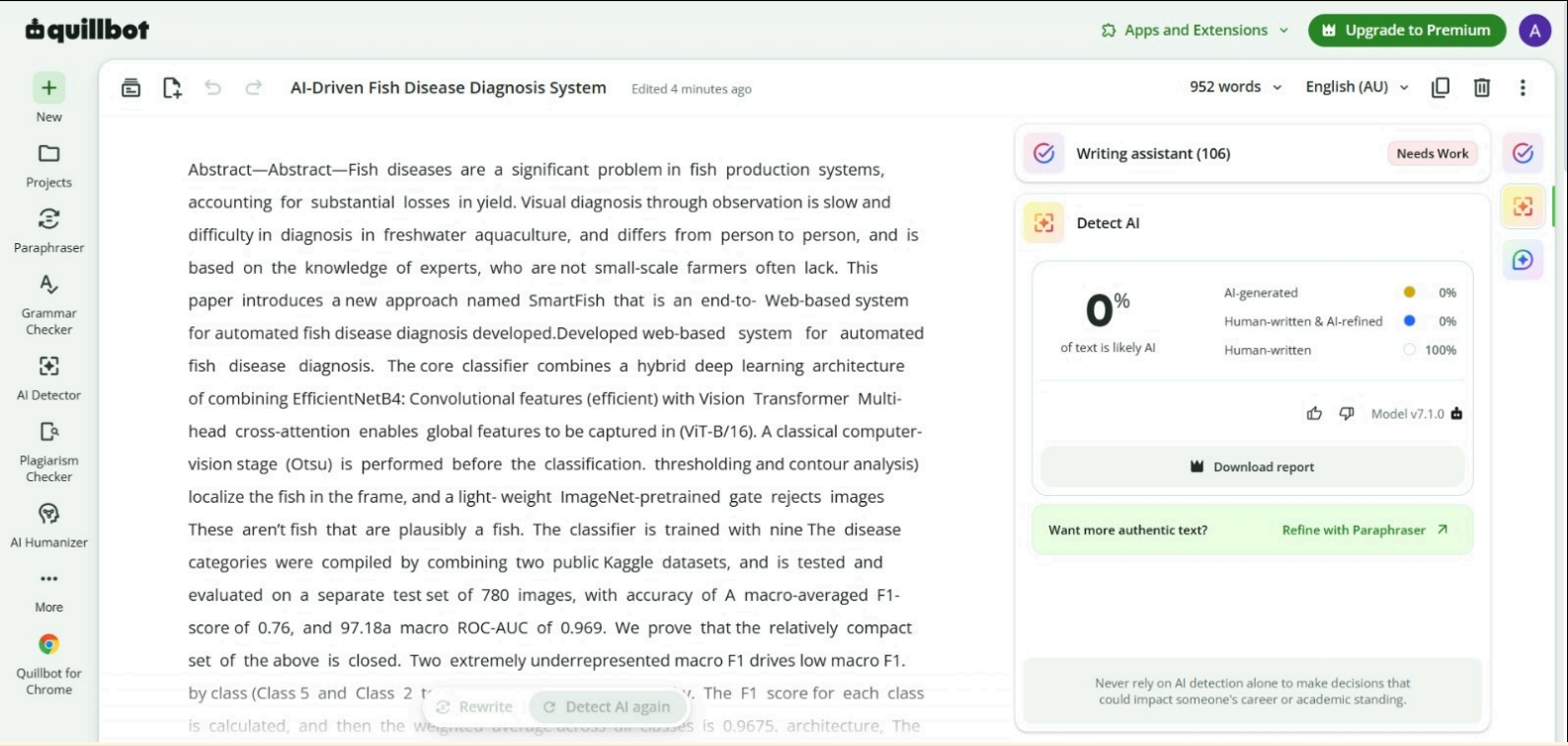
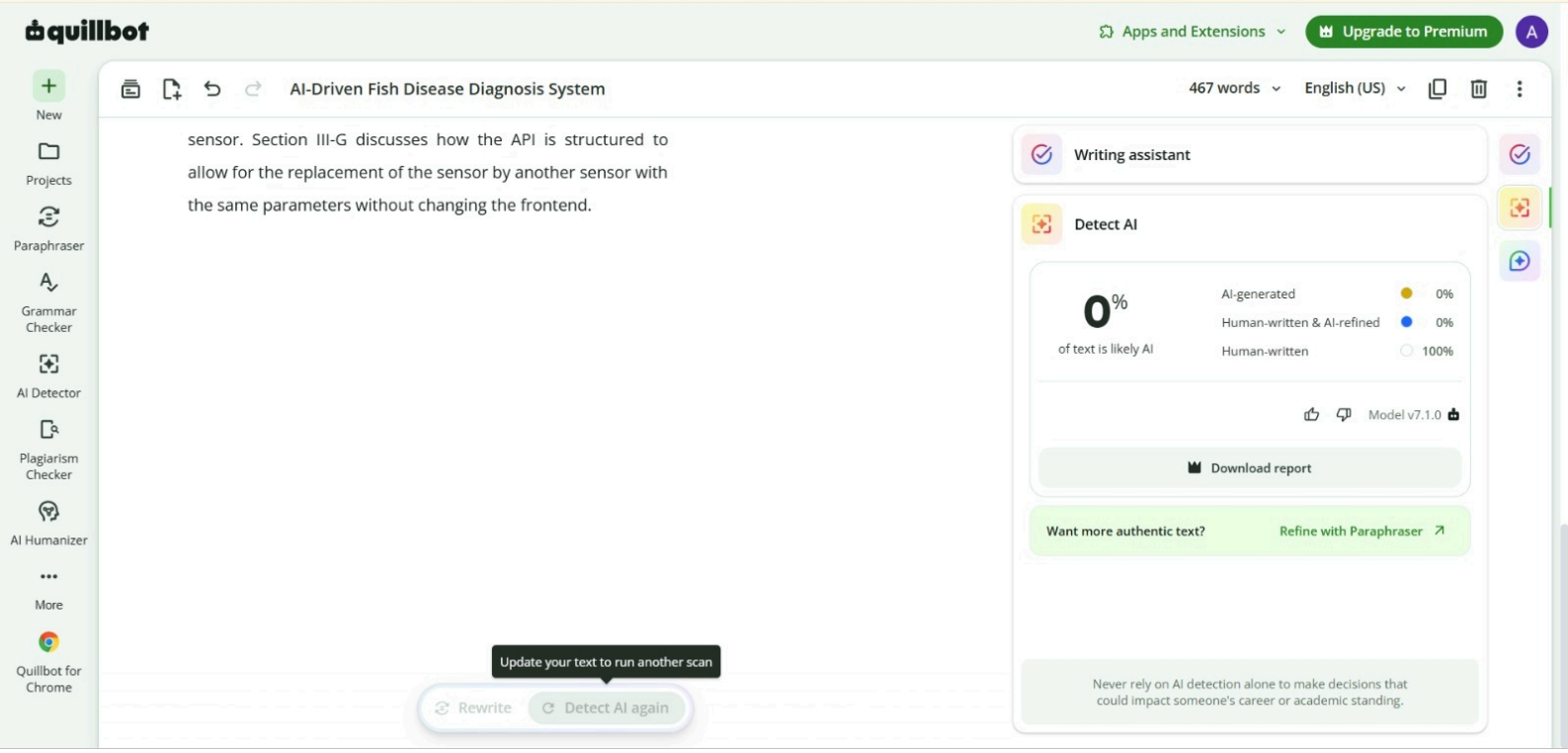

quillbot
Apps and Extensions
Upgrade to Premium
A
New
Projects
Paraphraser
Grammar Checker
AI Detector
Plagiarism Checker
AI Humanizer
More
Quillbot for Chrome
AI-Driven Fish Disease Diagnosis System
Edited 4 minutes ago
952 words
English (AU)
Abstract—Abstract—Fish diseases are a significant problem in fish production systems, accounting for substantial losses in yield. Visual diagnosis through observation is slow and difficulty in diagnosis in freshwater aquaculture, and differs from person to person, and is based on the knowledge of experts, who are not small-scale farmers often lack. This paper introduces a new approach named SmartFish that is an end-to- Web-based system for automated fish disease diagnosis developed.Developed web-based system for automated fish disease diagnosis. The core classifier combines a hybrid deep learning architecture of combining EfficientNetB4: Convolutional features (efficient) with Vision Transformer Multi-head cross-attention enables global features to be captured in (ViT-B/16). A classical computer-vision stage (Otsu) is performed before the classification. thresholding and contour analysis) localize the fish in the frame, and a light- weight ImageNet-pretrained gate rejects images These aren't fish that are plausibly a fish. The classifier is trained with nine The disease categories were compiled by combining two public Kaggle datasets, and is tested and evaluated on a separate test set of 780 images, with accuracy of A macro-averaged F1-score of 0.76, and 97.18a macro ROC-AUC of 0.969. We prove that the relatively compact set of the above is closed. Two extremely underrepresented macro F1 drives low macro F1. by class (Class 5 and Class 2 t
Rewrite
Detect AI again
is calculated, and then the
is 0.9675. architecture, The
Writing assistant (106)
Needs Work
Detect AI
0%
of text is likely AI
AI-generated 0%
Human-written & AI-refined 0%
Human-written 100%
Model v7.1.0
Download report
Want more authentic text?
Refine with Paraphraser
Never rely on AI detection alone to make decisions that could impact someone's career or academic standing.


quillbot
Apps and Extensions
Upgrade to Premium
A
New
Projects
Paraphraser
Grammar Checker
AI Detector
Plagiarism Checker
AI Humanizer
More
Quillbot for Chrome
AI-Driven Fish Disease Diagnosis System
467 words
English (US)
sensor. Section III-G discusses how the API is structured to allow for the replacement of the sensor by another sensor with the same parameters without changing the frontend.
Update your text to run another scan
Rewrite
Detect AI again
Writing assistant
Detect AI
0%
of text is likely AI
AI-generated 0%
Human-written & AI-refined 0%
Human-written 100%
Model v7.1.0
Download report
Want more authentic text?
Refine with Paraphraser
Never rely on AI detection alone to make decisions that could impact someone's career or academic standing.

quillbot
Apps and Extensions
Upgrade to Premium
A
New
Projects
Paraphraser
Grammar Checker
AI Detector
Plagiarism Checker
AI Humanizer
More
Quillbot for Chrome
AI-Driven Fish Disease Diagnosis System
1,116 words
English (US)

III. PROPOSED METHODOLOGY

A. Dataset curation

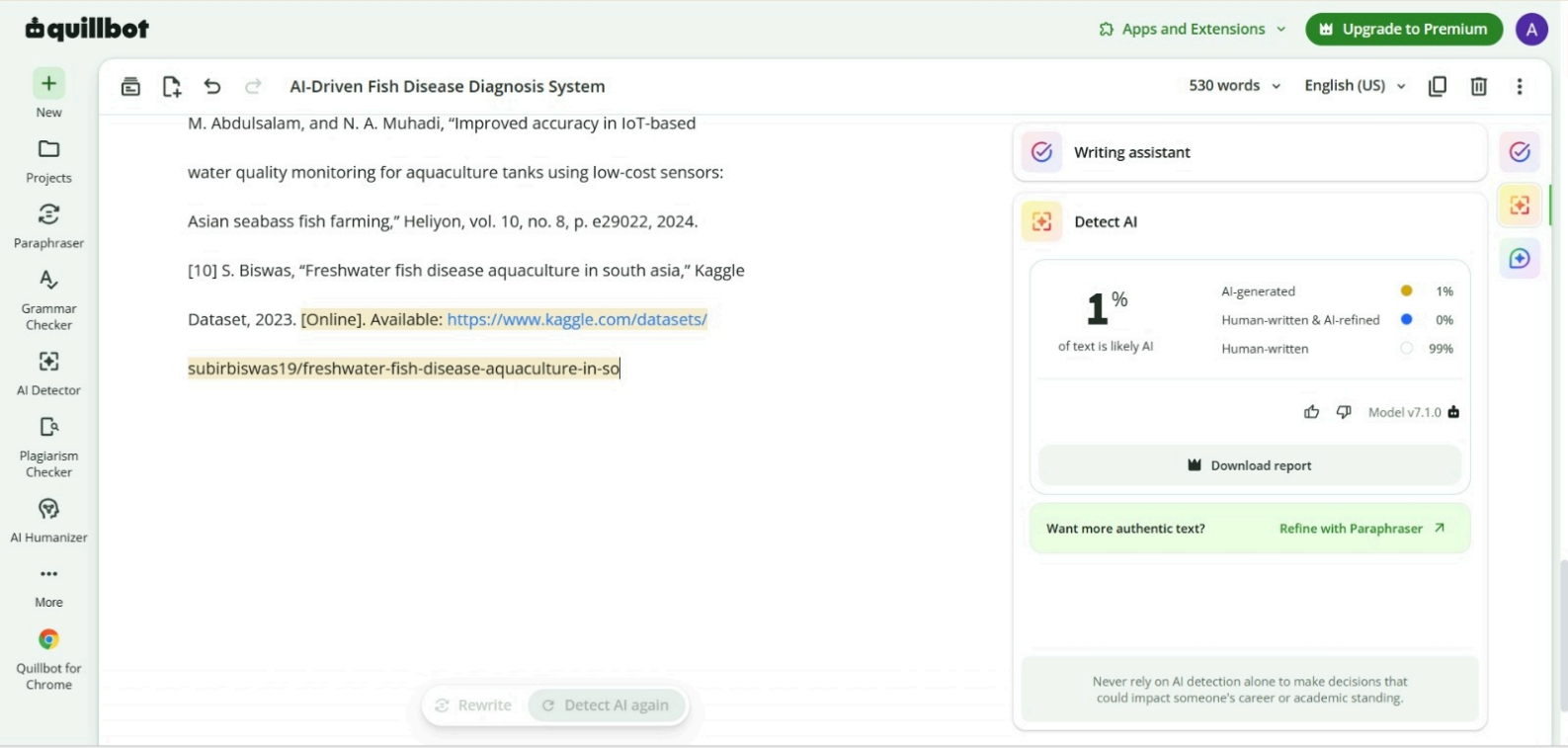
The poster's target taxonomy contains 11 categories of disease. This project combines two Kaggle datasets, Freshwater Fish Disease Aquaculture in South Asia [10] (contributing Bacterial Aeromoniasis, Bacterial Gill Disease, Bacterial Red Disease, Fungal Saprolegniasis, Parasitic Disease, Viral White Tail Disease, and Healthy Fish (~350 images/class)), and Fresh Water Fish Disease Dataset [11] (contributing Epizootic Ulcerative Syndrome (mapped to Ulcer_Disease) and Tail_And_Fin_Rot images (6–31 images/class, prior to splitting). The physical-injury ca

Rewrite
Detect AI again
Writing assistant
Detect AI
0%
of text is likely AI
AI-generated 0%
Human-written & AI-refined 0%
Human-written 100%
Model v7.1.0
Download report
Want more authentic text?
Refine with Paraphraser
Never rely on AI detection alone to make decisions that could impact someone's career or academic standing.

quillbot
Apps and Extensions
Upgrade to Premium
A
New
Projects
Paraphraser
Grammar Checker
AI Detector
Plagiarism Checker
AI Humanizer
More
Quillbot for Chrome
AI-Driven Fish Disease Diagnosis System
1,133 words
English (US)

physical sensors.

Update your text to run another scan
Rewrite
Detect AI again
Writing assistant
Detect AI
0%
of text is likely AI
AI-generated 0%
Human-written & AI-refined 0%
Human-written 100%
Model v7.1.0
Download report
Want more authentic text?
Refine with Paraphraser
Never rely on AI detection alone to make decisions that could impact someone's career or academic standing.


quillbot
Apps and Extensions
Upgrade to Premium
A
New
Projects
Paraphraser
Grammar Checker
AI Detector
Plagiarism Checker
AI Humanizer
More
Quillbot for Chrome
AI-Driven Fish Disease Diagnosis System
530 words
English (US)
M. Abdulsalam, and N. A. Muhadi, "Improved accuracy in IoT-based
water quality monitoring for aquaculture tanks using low-cost sensors:
Asian seabass fish farming," Heliyon, vol. 10, no. 8, p. e29022, 2024.
[10] S. Biswas, "Freshwater fish disease aquaculture in south asia," Kaggle
Dataset, 2023. [Online]. Available: https://www.kaggle.com/datasets/
subirbiswas19/freshwater-fish-disease-aquaculture-in-so
Writing assistant
Detect AI
1%
of text is likely AI
AI-generated 1%
Human-written & AI-refined 0%
Human-written 99%
Model v7.1.0
Download report
Want more authentic text?
Refine with Paraphraser
Never rely on AI detection alone to make decisions that could impact someone's career or academic standing.
Rewrite
Detect AI again